Biopsie osseuse guidée par scanner à comptage photonique et cartographie de l'œdème médullaire : ciblage des lésions occultes et lymphomes

Titre original : Photon-counting CT-guided bone biopsy with real-time bone marrow edema mapping

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1. Pertinence Clinique & Lien avec l'Agenda Interventionnel (Cochin)

Illustre le ciblage TDM de haute précision pour les biopsies osseuses (suspicion de récidive de lymphome B au genou, biopsie sacrée/pré-sacrée) et synoviales en HDJ Radio B.

2. Synthèse Clinique en Français

Résumé exécutif : Article de *European Radiology Experimental* (2026) montrant l'apport de l'imagerie spectrale / comptage photonique et de la corrélation IRM-TDM lors des biopsies osseuses sous scanner pour cibler les zones d'infiltration médullaire active (lymphome osseux, métastases médullaires) même en l'absence d'ostéolyse corticale franche.

Analyse détaillée : Les biopsies osseuses pour suspicion de récidive de lymphome B, d'infiltration hématologique ou de lésions sacrées/pré-sacrées présentent un défi particulier lorsque la trame trabéculaire paraît subnormale au scanner conventionnel ou remaniée par des traitements antérieurs. Cet article montre comment la fusion cognitive avec l'IRM (diffusion / T1 Gado) et l'analyse de densité médullaire permettent de diriger le trocart osseux exactement dans le foyer cellulaire actif, maximisant le rendement histologique et moléculaire tout en sécurisant les gestes sous anticoagulants.

3. Protocole Technique & Paramètres Opératoires

Corrélation précise avec les images clés PACS (IRM T1, STIR, Diffusion) • Pour les patients sous AOD (ex: Eliquis 2,5 mg ou 5 mg), arrêt à J-3, contrôle du bilan d'hémostase 48h avant et reprise > 6h après le geste • Trocart de biopsie osseuse coaxial 11G–13G : prélèvement de 3 à 4 carottes non fragmentées pour histologie et cytogénétique/FISH (lymphome).

4. Points Clés à Retenir pour la Pratique

- • Dans les suspicions de récidive de lymphome osseux, cibler la zone d'hypersignal diffusion / rehaussement IRM et éviter l'écrasement mécanique des carottes.
- • Pour les biopsies synoviales sous scanner (genou, suspicion de chondromatose synoviale ou SVN), utiliser un système coaxial 14G ciblant directement le nodule synovial du récessus.
- • Pour les masses pré-sacrées / sacrées (ex: suspicion de myélolipome ou chordome), l'abord trans-sacré ou para-sacré postérieur sous TDM évite toute effraction rectale ou plexique.
- • Protocole anticoagulants (Eliquis) pour biopsie osseuse profonde : arrêt J-3 et reprise > 6h post-procédure en l'absence de saignement.

5. Original PubMed Abstract (Texte Intégral de l'Abstract)

Objective Computed tomography-guided biopsies are needed to diagnose bone lesions, but can sometimes be challenging. We evaluated the feasibility and usefulness of photon-counting computed tomography (PCCT)-guided bone biopsies, focusing on real-time bone marrow oedema (BME) mapping to optimise diagnostic yield. Materials and methods This retrospective single-centre study included procedures performed

from September 2024 to May 2025 using a first-generation dual-source PCCT scanner with Quantum HD mode. Ten consecutive patients underwent PCCT-guided bone biopsy with real-time BME reconstructions. The reference standard was established using histopathology or microbiological confirmation when available; clinical and ≥ 3-month radiologic follow-up for nondiagnostic or discordant results. Statistical analysis included descriptive statistics, independent unpaired t-tests, and correlation analysis (SPSS v22.0, RStudio). Results Ten patients, five women and five men, aged 60.5 ± 13.5 years (mean ± standard deviation), were included in the final analysis. The overall diagnostic yield was 70% (7/10), with a diagnostic accuracy of 87.5% (7/8) for cases with a definitive reference standard. Final diagnoses comprised tumour bone metastases (n = 7, 70%), bone osteomyelitis (n = 1, 10%), and bone marrow deposition disease (n = 2, 20%). Mean radiation dose (dose-length product) was 644.5 ± 112.1 mGy·cm. Monoenergetic 70-keV imaging showed significant differences between mean HU values of lytic (42.6) and sclerotic lesions (476.2) ($p = 0.009$), with a strong negative correlation between lesion morphology (sclerotic versus lytic) and monoenergetic 70-keV attenuation values ($r = -0.84$; $p = 0.002$). Conclusion PCCT-guided bone biopsy with real-time BME mapping proved feasible and showed encouraging diagnostic performance in this small exploratory cohort. Larger validation studies are needed. Relevance statement By combining monoenergetic images and BME mapping, PCCT-guided bone biopsy improves lesion visualisation, operator confidence, procedural efficiency, and overall safety for diagnostic tissue sampling and active disease targeting. Key points Accurate targeting of active disease within complex bone lesions during CT-guided biopsy remains challenging sometimes. PCCT-guided biopsy with real-time BME mapping can enhance lesion targeting and procedural efficiency. PCCT-guided biopsy may improve safety, diagnostic accuracy, and operator confidence.

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6. Full-Text Sections (Europe PMC Open Access JATS XML)

Background

Photon-counting computed tomography (PCCT), first approved for clinical use by the USA Food and Drug Administration in 2021 [1], has marked a milestone in 21st-century radiology by incorporating a new acquisition technology. PCCT uses semiconductor-based detectors that directly convert x-ray photons into electrical signals, allowing for precise energy discrimination and the acquisition of pure multienergy data in a single scan. This contrasts with conventional spectral energy-integrating computed tomography (CT) detectors, which rely on scintillators to convert x-ray photons into visible light [2, 3]. PCCT equipment also eliminates interpixel septa within detectors, enabling ultrahigh (< 0.2 mm) spatial resolution acquisitions. In combination with energy thresholding technology, which removes electronic noise, this technology improves signal-to-noise ratios, increases radiation-use efficiency, and reduces dose [4].

Conventional CT-guided bone biopsy relies on anatomical landmarks and structural features (e.g., cortical destruction) to localise the target point. This approach, whilst adequate for some lesions, presents significant limitations when approaching complex osseous pathologies, particularly sclerotic or mixed lesions, where identifying viable diagnostic tissue remains challenging [5]. Traditional CT-guidance fails to differentiate between metabolically active regions and inactive or necrotic tissue, potentially leading to non-diagnostic sampling [5, 6]. Even though other imaging techniques allow it (magnetic resonance imaging or positron emission tomography-guided biopsy), they are not always available, technically possible or economically feasible.

Pathophysiologically, bone marrow oedema (BME) reflects the marrow’s response to a spectrum of insults, ranging from nonspecific reactive changes—such as trauma, microfractures, or transient vascular congestion—to active pathology, including infectious, inflammatory, and neoplastic conditions. In these scenarios, BME can be precisely assessed by spectral CT imaging [7–9]. Newly introduced PCCT pure multi-energy acquisition allows for real-time virtual non-calcium (VNCa) reconstructions by suppressing calcium attenuation whilst enhancing the visualisation of bone marrow, generating a colour-coded BME map that can be used to assess intramedullary pathologies [10, 11]. This guidance tool might potentially increase diagnostic precision, reduce procedure time, and boost the operator’s targeting confidence. However, real-time mapping

of BME during the biopsy itself remains an unexplored frontier.

This study aims to evaluate the clinical usefulness and feasibility of PCCT-guided bone biopsies, specifically by integrating real-time BME mapping via VNCa reconstructions to optimise diagnostic yield and procedural efficiency.

Materials and methods

This retrospective study received approval from the institutional review board (CEIC 2025.016). From September 2024 to May 2025, a total of ten consecutive patients were referred to the Radiology Department to undergo PCCT-guided bone biopsy of suspected bone lesions identified on prior imaging techniques. Informed consent for undergoing the procedure was obtained from all participants. Patients were included if lesions were accessible for safe intervention, tolerated positioning, and had no bleeding disorders, were not taking anticoagulants, and had no active infections. Only one lesion per patient was included for analysis, and all outcomes (diagnostic yield, diagnostic accuracy, imaging metrics, and radiation dose) were calculated on a per-patient basis.

The bone biopsies were performed using a first-generation dual-source PCCT scanner (NAEOTOM Alpha, Siemens Healthineers). Scans were conducted in Quantum HD scan mode with a collimation of 96 × 0.2 mm. The system generated spectral imaging results, including single-photon processing, with standard reconstructions performed at slice thicknesses as low as 0.4 mm using kernels of up to 80. Additionally, Quantum HD reconstructions achieved a slice thickness of 0.2 mm. Systems A and B operated at different kilovolt peak (kVp) and threshold settings, with a thick tin filter and a tube voltage of 150 kV. The specific kVp configurations included 70/Sn150 kVp and 90/Sn150 kVp. Only the smallest collimation was employed to ensure ultrahigh resolution imaging, and only noncontrast scan protocols were used for this study. Reconstruction techniques included T3D reconstructions at full ultrahigh resolution, as well as additional spectral reconstructions for monoenergetic images, bone marrow analysis, and single-photon processing (Table 1).Table 1Acquisition and reconstruction parametersScanner modelNAETOM alphaSource modeDual sourceTube voltage70/Sn150Pitch0.6Collimation96 × 0.2 mmQIRNone*Pixel spacing (in x- and y-axis)0.5 mmSeriesBoneSoft tissueBME mapping Slice thickness0.6 mm0.6 mm2 mm Slice interval0.4 mm0.4 mm2 mm Reconstruction filterBr80Br40Qr40 Matrix size768 × 768512 × 512512 × 512BME Bone marrow oedema, QIR Quantum iterative reconstruction* To enable BME map processing, data acquisition must be performed in a specific vendor-defined multi-energy mode. This acquisition mode imposes certain technical limitations, such as the unavailability of QIR implementation

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These advanced imaging techniques provided detailed visualisation of the suspect bone lesions, facilitating precise CT-guided biopsy procedures. In the evaluation of bone lesions by PCCT prior to biopsy, key parameters were evaluated to characterise the lesions and improve procedure planning. The analysis identified the number and anatomic location of lesions, predominantly in the iliac, sacral and femoral bones. Lesions were morphologically classified as lytic or sclerotic, measured in millimetres. For each lesion, HU values were recorded at a mean energy level of 70 keV, which closely approximates the attenuation characteristics of 120-kVp imaging while providing improved image quality through reduced noise and artefact levels, thereby enhancing assessment of tissue density and composition [12]. In addition, the presence of oedema was assessed on BME maps, suggesting lesion activity. These findings were instrumental not only in characterising the lesions but also in determining optimal puncture sites, thus improving the diagnostic yield of the biopsy procedure.

For intraprocedural bone-marrow analysis, BME maps were generated using a vendor-specific three-material decomposition model (water, fat, and calcium) applied to the spectral dataset. VNCa processing suppressed calcium attenuation to enhance marrow attenuation differences. The colour scale was vendor-calibrated, with yellow-green values representing increased VNCa attenuation (interpreted as pathological oedema) and blue-violet values representing normal marrow. BME maps were reconstructed using a 2-mm slice thickness and 2-mm increment with the Qr40 spectral kernel, whereas standard high-resolution diagnostic series were reconstructed at 0.6 mm (Br80/Br40) and ultra-high-resolution reconstructions at 0.2–0.4 mm. These parameters are now fully aligned with Table 1.

For each procedure, dose accumulation parameters were recorded in detail. The dose-length product (DLP) reflected the sum of all acquisitions performed during the intervention, including the planning scan, incremental checks during needle advancement, and the final post-biopsy control scan. All BME/VNCa reconstructions were derived from the same spectral dataset and did not require additional acquisitions. For each series, the scan range, collimation, and kVp/threshold settings (70/Sn150 or 90/Sn150 kVp) were documented.

Results

A total of ten patients referred for a PCCT-guided bone biopsy, aged 60.5 ± 13.5 years (mean ± standard deviation), five women and five men, were enroled in the study (Fig. 1). Each patient contributed a single target lesion to the analysis; therefore, diagnostic yield, accuracy, and morphologic classification (lytic versus sclerotic) were assessed per patient.Fig. 1Study the inclusion process of PCCT-guided bone biopsies with exclusion criteria. DLP, Dose-length product; PCCT, Photon-counting computed tomography

Study the inclusion process of PCCT-guided bone biopsies with exclusion criteria. DLP, Dose-length product; PCCT, Photon-counting computed tomography

The overall diagnostic yield for all bone lesions examined in this study stood at 70% (7/10), with an overall diagnostic accuracy of 87.5% (7/8).

There were 7 tumour bone metastases, 1 bone osteomyelitis, and 2 bone marrow infiltrative disease (Table 2). A per-patient reference standard was available in all cases: histopathology in eight patients and clinical/radiologic follow-up at ≥ 3 months in two patients. One nondiagnostic biopsy corresponded to a lesion subsequently confirmed as metastatic on follow-up (false negative), whereas another nondiagnostic biopsy remained stable on imaging, constituting a true negative. The corresponding 2 × 2 contingency table and 95% confidence intervals for diagnostic yield and accuracy are provided in the Supplementary material.Table 2Per-lesion demographic and clinical dataAge (years)60.5 ± 13.5Sex Male5 (50%) Female5 (50%)Location Axial5 (50%) Appendicular5 (50%)Bone matrix Lytic5 (50%) Sclerotic5 (50%)Procedural average time (min)48.6 ± 8.6DLP (mGy·cm)644.5 ± 112.1Final diagnosis Malignant disease7 (70%) Osteomyelitis1 (10%) Deposition disease2 (20%)Data are given as mean ± standard deviation or absolute values (percentage)DLP Dose-length product

Per-lesion demographic and clinical data

Data are given as mean ± standard deviation or absolute values (percentage)

The mean operation time was 48.6 ± 8.6 min, ranging from 40 to 66 min per bone biopsy procedure.

A mean DLP of 644.5 ± 112.1 mGy·cm was found for PCCT-guided bone biopsy. The mean DLP for lytic lesions was 614.6 mGy·cm, while for sclerotic lesions, it was 674.4 mGy·cm.

Discussion

In this study, we highlighted the potential of PCCT as a guiding technique for bone biopsy procedures, demonstrating its ability to reduce radiation exposure significantly. Specifically, from our study, PCCT achieved an average DLP of 644.5 mGy·cm for CT-guided bone biopsy, which is lower compared to the reported DLP value of 733 mGy·cm using conventional CT scans [14] and 784 mGy·cm with dual-energy CT techniques [15, 16]. In this cohort, all metrics were evaluated per patient to avoid spectrum inflation from multiple lesions in the same individual and to provide a clearer interpretation of targeting performance.

PCCT represents a significant advancement in imaging technology, offering unique features that improve procedural accuracy and efficiency [11]. Similar to dual-energy CT, PCCT utilises spectral imaging capabilities to differentiate materials based on their atomic number and electron density. This enables the subtraction of calcium from trabecular bone, facilitating the identification of BME [17]. Such imaging provides diagnostic information comparable to magnetic resonance imaging, particularly T2-weighted fat-suppressed images, and is highly effective in detecting BME in pathological conditions such as neoplasms [18].

Notably, PCCT enables real-time imaging of intralesional bone oedema during biopsy procedures, facilitating precise localisation of the area of interest. This capability allows for dynamic guidance of the biopsy system as it advances toward the targeted region, enhancing the accuracy of the procedure, particularly in cases where precise lesion targeting is paramount.

In addition to improving accuracy, PCCT positively impacts procedural efficiency and workflow. By enabling real-time visualisation of bone oedema, PCCT reduces the need for repeated scans and adjustments, therefore streamlining the process and minimising procedural time. In fact, the operation time was 49 min, comparable to previous reports [19]. This not only improves patient comfort and reduces radiation exposure but also optimises clinical workflow, allowing for a greater number of procedures to be performed within a given timeframe.

Previous studies have noted that conventional CT-guided biopsies often exhibit lower diagnostic yield and a significant false-negative rate [20], particularly in sclerotic lesions [16]. These limitations frequently arise from the inability of conventional CT to reliably distinguish between viable tumour tissue and reactive sclerosis. In contrast, this study has shown promising results for monoenergetic 70 keV imaging in PCCT, with a statistically significant difference observed between lytic and sclerotic lesions ($p = 0.009$). Furthermore, a strong negative correlation ($r = -0.84$; $p = 0.002$) was found between lesion morphology (binary coding: sclerotic = 0; lytic = 1) and monoenergetic 70-keV attenuation values. These findings underscore the potential of monoenergetic 70 keV imaging to enhance the differentiation between lytic and sclerotic lesions, thereby improving diagnostic accuracy and visualisation of bone matrix. In the present study, diagnostic yield reflects only the proportion of biopsies providing a definitive histopathologic diagnosis, whereas diagnostic accuracy reflects agreement with the per-patient reference standard. Importantly, final diagnoses in nondiagnostic cases were established through structured clinical and radiologic follow-up, preventing misclassification. Although the pathologist was not formally blinded to clinical information, biopsy interpretation was performed independently, which limits verification bias.

Additionally, PCCT improves the visualisation of trabecular bone microarchitecture and enhances diagnostic precision, particularly in identifying and characterising lytic lesions, with slice thicknesses as thin as 0.2 mm, surpassing the capabilities of conventional energy integrating detector CT systems, which are constrained by larger detector elements and higher noise levels [21]. This improved resolution is particularly beneficial for identifying small or early-stage lytic lesions [10], which are often challenging to detect with conventional imaging methods [22]. It also facilitates more accurate targeting during procedures such as bone biopsies, thereby reducing the risk of sampling errors and improving diagnostic reliability.

This study has several limitations that should be acknowledged. First, it is a single-centre study with a limited number of patients, which restricts the ability to conduct a comprehensive analysis of subgroups, particularly in assessing lytic and sclerotic bone lesions. Second, the qualitative analysis of real-time bone oedema is inherently subjective and may vary between different observers, potentially affecting the consistency of the findings. Third, when a bone lesion is located outside the centre of the CT scan, repeated preliminary acquisitions are often necessary to ensure the safety and accuracy of the multi-energy information collected. Fourth, the reconstruction settings for PCCT, including resolution, kernel, denoising, and radiation dose, can vary depending on the affected body part, potentially influencing diagnostic imaging features. Additionally, the long-term clinical outcomes associated with PCCT-guided bone biopsies, particularly regarding their impact on patient management and overall prognosis, warrant further exploration in future research.

Despite these limitations, the overall findings highlight the clinical potential of this technique. PCCT-guided bone biopsy with real-time BME mapping was feasible and provided promising guidance capabilities in this

small exploratory cohort. The technique enabled improved visualisation of intramedullary changes and yielded encouraging diagnostic performance metrics. While preliminary, these results support further evaluation in larger, controlled studies to better define its clinical value and comparative effectiveness.